

IMO Training 2018–19

Phase 2 (Level II)

Games and Strategies

2018/10/06

The following should be assumed unless the content suggests otherwise:

- Ann and Ben are playing a sequential game with Ann going first.
- If a problem involves placing chess pieces on a chessboard, each piece should be placed within a cell and each cell can accommodate at most one piece.

Examples

1. Ann and Ben take turns to place 1-dollar coins within a square board and the coins must not overlap each other. The first player failing to do so loses. Who has a winning strategy?

2. (Modified from IMO 2018)

Ann and Ben take turns to put chess pieces on a 20×20 chessboard. Ann is placing non-attacking knights while Ben is placing dummy pieces. How many knights can Ann guarantee to place?

3. (Chomp)

Ann and Ben take turns to put chess pieces on an $m \times n$ chessboard whose cells are centred at (x, y) where $x = 1, 2, \dots, m$ and $y = 1, 2, \dots, n$. Each time the player must place a chess piece in an empty cell, with the requirement that if there is a chess piece in (a, b) , one cannot place a chess piece in (c, d) with $c \geq a$ and $d \geq b$. The player who places a chess piece at $(1, 1)$ loses.

(a) Show that Ann has a winning strategy if $m = n > 1$.

(b) Who has a winning strategy if $m \neq n$?

4. (Nim)

There are three piles of candies. Ann and Ben take turns to eat (at least one) candies from one of the piles each time. The first player failing to do so loses. Who has a winning strategy?

Exercise

5. There are two piles of candies. Each time, Ann and Ben take turns to either eat two candies from the same pile or transfer one candy from the first pile to the second. The first player failing to do so loses. Who has a winning strategy?

6. (Modified from Israel 1995)

Ann and Ben play a special version of tic-tac-toe on an infinite chessboard. One wins by marking a row or column of five consecutive cells. Show that Ann has no winning strategy.

7. (Modified from Ireland 1996)

In a game, x red and y blue chess pieces have been placed on a 5×9 chessboard at the beginning. Ann and Ben take turns to do the following:

- During Ann's turn, each red piece is moved vertically while each blue piece is moved horizontally to an adjacent cell.
- During Ben's turn, each red piece is moved horizontally while each blue piece is moved vertically to an adjacent cell.

- (a) If $x + y = 33$, show that either Ann or Ben will eventually be unable to complete the task.
(b) Is the conclusion in (a) still true if 33 is replaced by 32?

8. (Modified from Russia 2000)

There are two piles of candies. Each time, Ann and Ben take turns to eat (at least one) candies from one of the piles (say, pile X), and then transfer the same number of candies from pile X to the other pile. The first player failing to do so loses. Who has a winning strategy?